First-order Convergence for Finite-Domain CSPs

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Joint work with Colin Jahel

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- Classes of the form CSP(D) for finite digraph D.

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- It is homogeneous: all isomorphisms between finite substructures can be extended to automorphisms of $\mathbb{G}$.
- $T^{\text{as}}_{\mathcal{G}}$ is logically equivalent to a set of extension axioms, i.e., sentences of the form $\forall x_1, \ldots, x_n \exists y. \psi$ where ψ is a conjunction of atomic formulas and negated atomic formulas.

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2. When can we show pseudofiniteness via the probabilistic method?

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Therefore,

$$T_{\mathcal{C}}^{\mathrm{as}} \neq \mathrm{Th}(\mathbb{C}) := \{\varphi \mid \mathbb{C} \models \varphi\}$$

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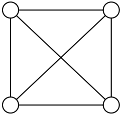

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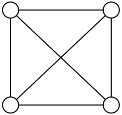

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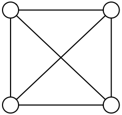

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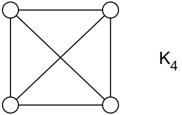

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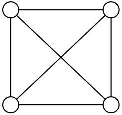

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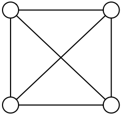

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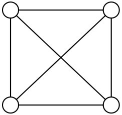

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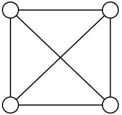

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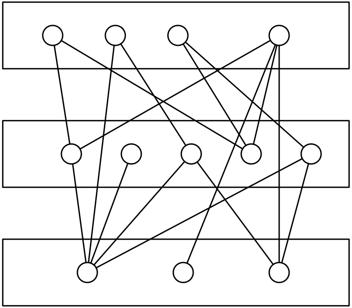

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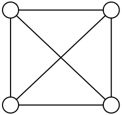

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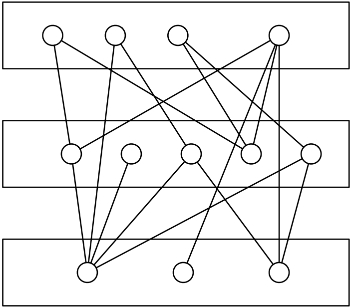

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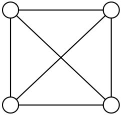

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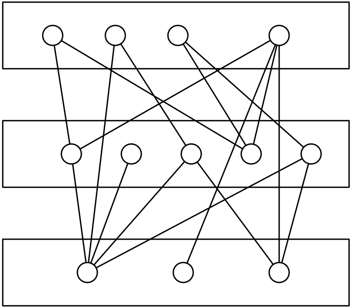

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- D. Kühn, D. Osthus, T. Townsend, Y. Zhao'17:
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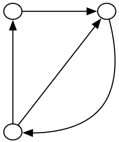

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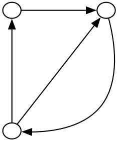

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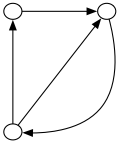

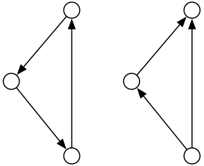

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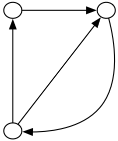

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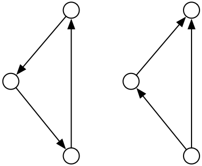

- ChatGPT 5.6 Sol: There exists a finite oriented graph D such that $\mathrm{CSP}(D)$ does not have first-order convergence.

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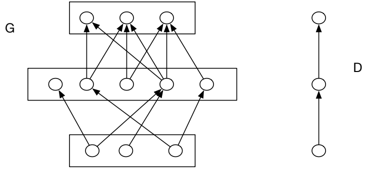

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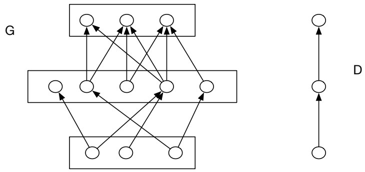

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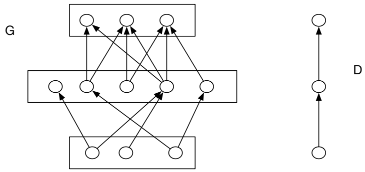

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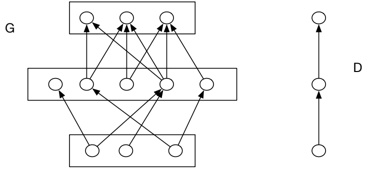

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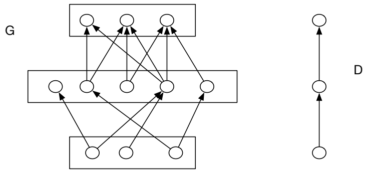

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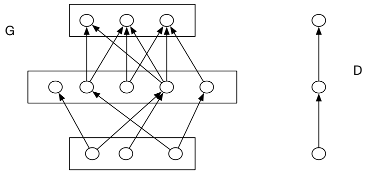

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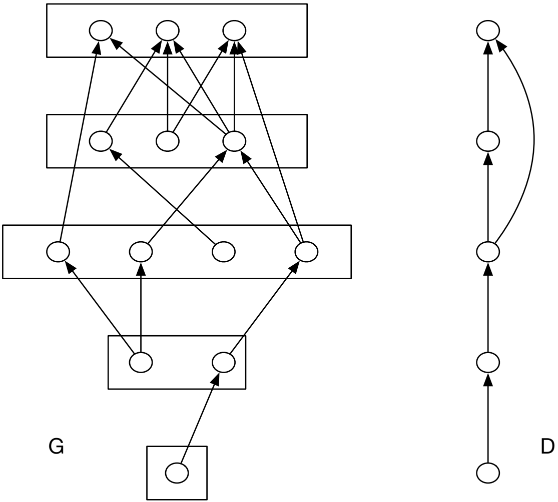
An Example



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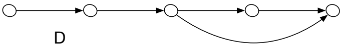

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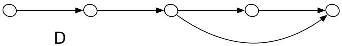

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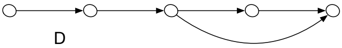

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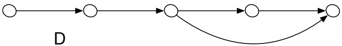

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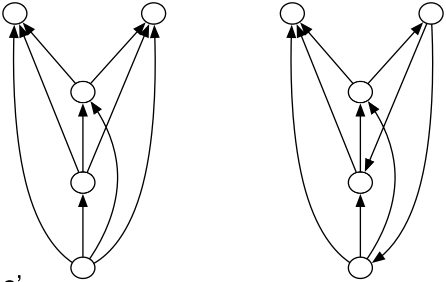

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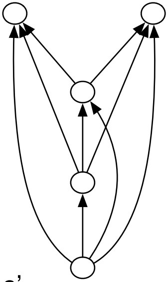

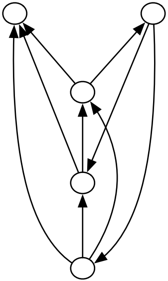

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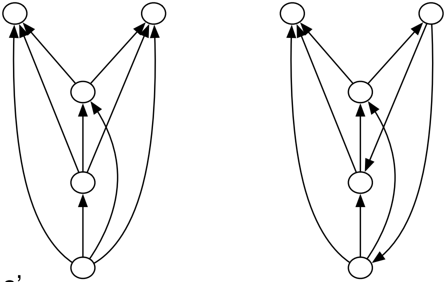

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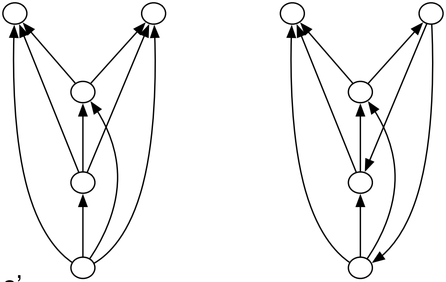

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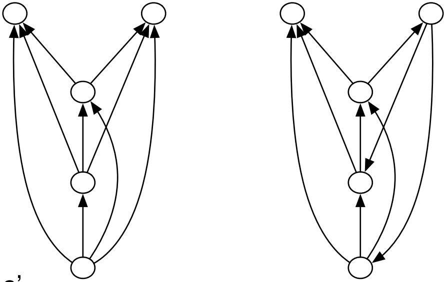
The Support of Density Functions



Definition. A blow-up of a digraph G:
replace each vertex of G by many 'copies'.

Example. Blow-up of K_ℓ is complete ℓ-partite graph $K_{k_1,\dots,k_\ell}$.

Lemma. Let D be an oriented graph and let δ be a density function. Then

- the subgraph of D induced by the support of δ is an orientation of a blow-up of K_k;
- the sum over $\delta(u)$ for all copies u of the same vertex of K_k equals $1/k$.

Random D-colored digraphs

$D = (V, E)$: finite directed graph

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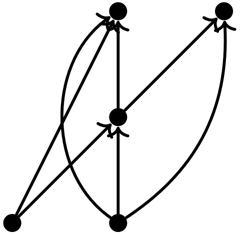

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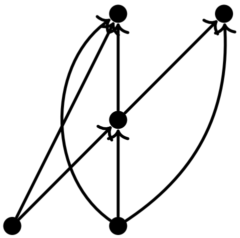

Theorem. For every $\epsilon > 0$,
in a random D-colored digraph (G, f),
a.a.s. there exists a density function δ such that
for every v in the support of δ:

$$\lfloor n(\delta(v) - \epsilon) \rfloor \leq |f^{-1}(v)|.$$

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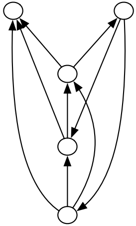

Lemma. Let D be a finite oriented digraph and $\gamma \in \mathcal{O}$. Then

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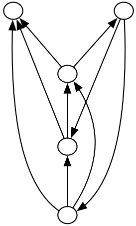

Lemma. Let D be a finite oriented digraph and $\gamma \in \mathcal{O}$. Then

- $\{v \in V(D) \mid \gamma(v) > 0\}$ is an inclusion-wise maximal orientation of a blow-up of K_ℓ.
- If A_i is the set of copies of vertex i in the blow-up of K_ℓ, then

$$\gamma(u) = \begin{cases} \frac{1}{\ell |A_i|} & \text{if } u \in A_i \\ 0 & \text{otherwise.} \end{cases}$$

The random theory of D-colored digraphs

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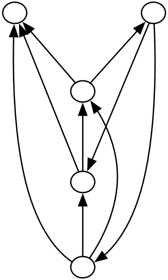

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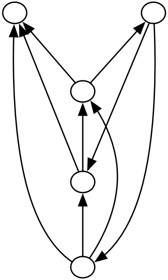

Proposition. The almost-sure theory
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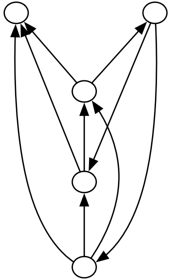

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Proof: Suffices to verify extension axioms.

Getting rid of the colors 1

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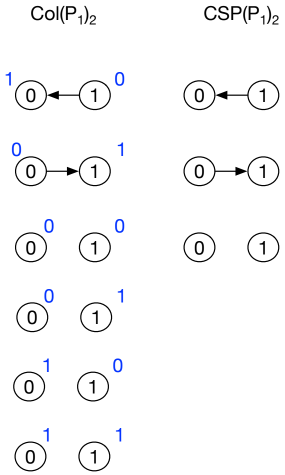

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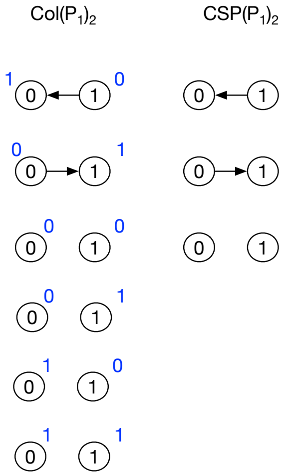

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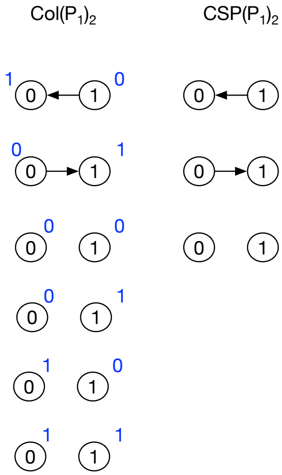

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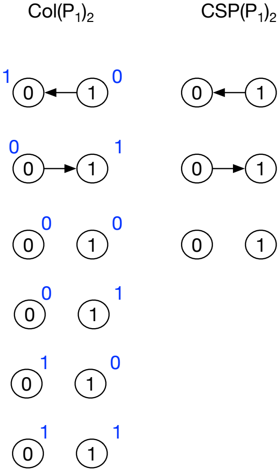

Getting rid of the colors 2

Getting rid of the colors 2

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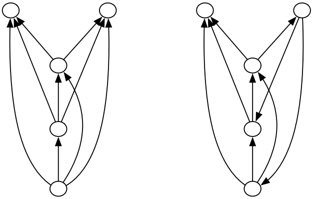

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The almost-sure first-order theory of CSP(D)
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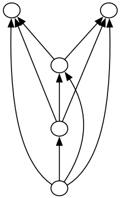
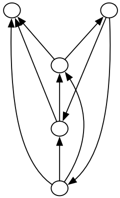

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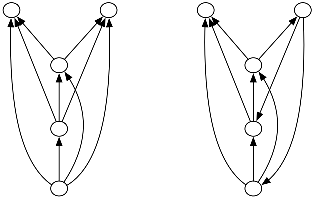

Theorem.

Let D be an oriented graph with a unique subgraph S which is an orientation of a $K_{k_1,\dots,k_\ell}$ such that $(\ell, \prod_{i=1}^{\ell} k_i)$ is lexicographically maximal

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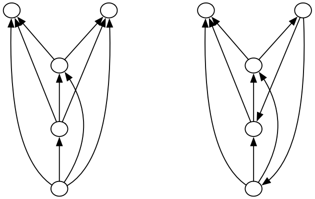

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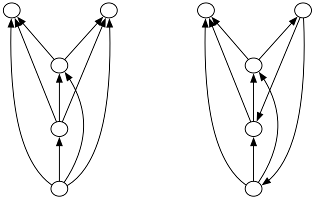

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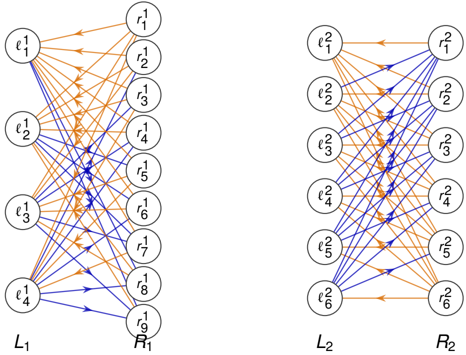
C_1: an orientation of $K_{4,9}$ **C_2: an orientation of $K_{6,6}$**



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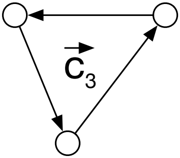
AI disclosure: used ChatGPT 5.6 Sol to find this example.
Argument human-verified.

Open Problems



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1. Does the class Hom-Forb$(\vec{C}_3)$ have a 0-1 law?

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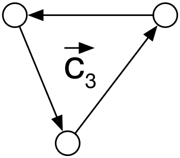

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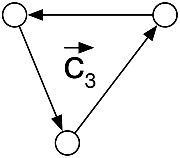

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3. Is the theory of the homogeneous universal triangle-free graph pseudofinite?